

Tan Dai Ngo

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EDUCATION

University of Victoria | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Courses:** Algorithms and Data Models, Systems for Massive Datasets, Information Theory. **GPA:** 8.78 / 9.00.

University of Chicago | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Courses:** Distributed Systems, Databases, Applied Software Engineering, Networks.

University of Washington | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics. **GPA:** 3.57 / 4.00.

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot.
- Tuned Cassandra queries to keep response times under 3 seconds across the Roaming Business System services.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and shipped fixes via Jenkins across 50+ releases.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.
- Explained technical material to students with no prior background, adjusting the pitch in real time.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, PyTorch, XGBoost *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six supervised classifiers on 206,000 CIC IoT-DIAD 2024 flows with preprocessing fit on training only.
- Selected binary XGBoost under a 1% false positive budget, reaching 99.5% recall at 0.94% FPR and 0.989 PR-AUC.
- Designed a four condition holdout that exposed capture session leakage, dropping PR-AUC from 0.981 to 0.125.

Ocean Data ML Platform with RAG | Python, Spark, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Ingested 10.8M NOAA and Ocean Networks Canada records through Bronze and Silver into a 3.69M row Gold table.
- Served model inference and plain English retrieval behind FastAPI with Qdrant and Ollama, scoring hit@k 0.90.
- Containerized the FastAPI service and vector store with Docker Compose so local runs matched deployment.

Anomaly Detection at Scale on Species Data | Python, PySpark, scikit-learn *UVic, team of 3 | Mar - Apr 2026*

- Engineered Isolation Forest from the published algorithm rather than importing a library, then validated it.
- Verified linear runtime growth on 1,093,203 eBird records using stratified samples across eight quantile bins.
- Identified a geographic form of swamping and proposed rank transformation and density aware subsampling.

Automated Crypto Trading Bot | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine that buys 1% dips and sells on 2% rises inside rolling 60 minute windows.
- Added a trend strategy that buys only when the 8 period EMA is above the 20 and price is above the 200 EMA.
- Backtested both on four months of polled data, then deployed to AWS EC2 with per market caps from a CSV config.

TECHNICAL SKILLS

Languages & Programming: Python, Java, object oriented programming, scripting

Machine Learning & Evaluation: PyTorch, XGBoost, scikit-learn, pandas, NumPy, Optuna, MLflow, classification

Data, Retrieval & Messaging: Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, Kafka

Backend, Cloud & CI/CD: Spring Boot, FastAPI, REST, Docker, AWS (EC2, S3, Lambda, SNS), Jenkins deployments

Engineering Practice: pytest, API testing, model evaluation, experiment tracking, code review, Postman, Jira